

RAMSTEIN, Germany

WMO: 106140

Lat: 49.437N

Lon: 7.600E

Elev: 238

StdP: 98.50

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.2	-12.4	1.3	-9.2	-9.8	1.7	-5.9	11.2	8.1	9.9	8.1	1.7	90	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	31.5	20.3	29.2	19.4	27.2	18.7	21.7	28.3	20.6	27.1	19.6	25.5	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.0	23.5	18.6	13.8	22.2	17.7	13.1	21.6	64.1	28.7	60.1	27.4	56.9	25.6	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.7	6.8	-14.5	34.8	3.5	1.9	-17.0	36.2	-19.1	37.3	-21.0	38.4	-23.6	39.8		
			-14.6	23.6	3.5	1.9	-17.2	25.0	-19.2	26.0	-21.2	27.1	-23.7	28.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	1.5	2.5	5.5	9.4	13.3	16.8	18.9	18.3	14.1	10.2	5.6	2.5
	DBStd	7.19	4.69	4.26	3.81	3.67	3.53	3.52	3.23	3.22	3.15	3.74	3.89	4.48
	HDD10.0	1109	266	211	145	56	10	1	0	0	4	45	138	234
	HDD18.3	3195	523	444	397	269	159	70	33	40	133	253	382	491
	CDD10.0	1079	1	1	6	36	114	206	276	258	125	50	7	2
	CDD18.3	124	0	0	0	0	4	25	50	39	4	1	0	0
	CDH23.3	1717	0	0	0	17	98	351	642	523	83	3	0	0
	CDH26.7	568	0	0	0	1	18	103	235	194	16	0	0	0
Wind	WSAvg	2.6	3.2	3.0	3.0	2.6	2.6	2.4	2.4	2.2	2.2	2.4	2.5	3.1
Precipitation	PrecAvg	878	80	76	71	53	68	65	68	71	68	84	80	93
	PrecMax	1162	188	189	182	128	132	153	183	188	123	231	171	271
	PrecMin	613	13	11	12	2	12	23	32	4	21	19	23	51
	PrecStd	150	42	46	44	30	32	33	32	41	31	45	38	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	14.8	20.1	25.8	29.0	32.2	34.2	35.2	29.0	23.2	16.3	13.1
		MCWB	9.9	9.7	11.8	14.9	18.4	21.0	20.9	20.9	19.5	18.6	12.7	10.4
	2%	DB	11.1	12.1	17.1	22.9	26.0	29.7	31.8	31.1	25.8	20.1	14.2	11.8
		MCWB	9.3	8.5	10.5	14.2	17.4	19.7	20.4	20.2	18.7	16.1	11.7	10.0
	5%	DB	9.2	10.2	14.6	20.1	23.9	27.2	29.2	28.5	23.1	18.1	12.9	10.2
		MCWB	7.7	7.9	9.6	13.0	16.1	18.9	19.3	19.4	17.3	14.7	11.0	8.8
	10%	DB	8.0	8.9	12.1	17.7	21.3	25.1	27.0	26.1	20.9	16.1	11.2	9.0
		MCWB	6.7	7.2	9.0	11.6	14.9	17.8	18.7	18.6	16.1	13.6	9.5	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.9	13.2	16.7	20.2	23.0	22.9	23.7	22.1	19.5	14.1	11.7
		MCDB	12.0	13.0	17.5	22.9	26.3	29.8	30.6	27.5	26.0	21.2	15.3	12.4
	2%	WB	9.6	9.4	11.7	14.7	18.4	20.9	21.3	21.4	19.2	16.7	12.3	10.3
		MCDB	10.9	11.1	15.3	21.3	24.1	26.7	29.0	28.2	23.7	19.3	13.8	11.3
	5%	WB	8.0	8.3	10.4	13.2	16.9	19.6	20.3	20.2	17.8	15.2	11.0	9.1
		MCDB	9.3	10.0	13.3	18.6	22.1	25.5	27.0	26.9	22.3	17.5	12.7	10.1
	10%	WB	6.6	7.1	9.4	12.1	15.6	18.5	19.4	19.3	16.5	13.9	9.8	7.9
		MCDB	7.7	8.6	12.1	16.9	20.1	24.0	25.3	24.9	20.2	16.0	11.2	8.9
Mean Daily Temperature Range	5% DB	MDBR	5.6	7.5	10.1	12.2	12.0	12.2	12.4	12.5	11.7	9.6	6.1	5.2
		MCDBR	6.1	10.5	15.7	18.0	17.0	16.5	17.5	17.6	16.2	12.6	8.1	6.2
		MCWBR	5.0	7.7	9.9	9.9	8.5	7.8	7.2	7.6	9.2	8.7	6.0	5.1
	5% WB	MCDBR	5.8	8.9	12.3	15.4	14.7	14.7	15.0	15.2	14.2	11.8	7.5	5.7
		MCWBR	5.1	7.1	8.4	8.9	7.9	7.5	6.9	7.6	8.7	8.7	5.9	5.1
Clear-Sky Solar Irradiance	taub		0.318	0.346	0.378	0.406	0.414	0.417	0.415	0.411	0.384	0.369	0.340	0.316
	taud		2.398	2.343	2.270	2.232	2.240	2.263	2.259	2.299	2.348	2.378	2.395	2.395
	Ebn at Noon		729	790	823	838	846	844	839	825	809	752	690	674
	Edh at Noon		66	87	111	128	132	130	129	118	102	84	66	58
All-Sky Solar Radiation	RadAvg		0.82	1.56	2.76	4.29	5.07	5.67	5.37	4.60	3.46	2.00	0.93	0.65
	RadStd		0.12	0.29	0.42	0.55	0.54	0.49	0.55	0.47	0.41	0.18	0.10	0.10

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page